



College of
EDUCATION

PRINCIPAL COMPONENTS ANALYSIS, EXPLORATORY FACTOR ANALYSIS, AND CONDUCTING EXPLORATORY ANALYSES WITH CFA

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- Methods for exploratory factor analysis (EFA)
 - Principal Components-based
 - Maximum Likelihood-based Exploratory Factor Analysis
 - Exploratory Structural Equation Modeling

- Comparisons of CFA and EFA

- How to do exploratory analyses with CFA
 - Structure of no items known
 - Structure of some items known

- Exploratory analyses attempt to discover hidden structure in data with little to no user input
 - Aside from the selection of analysis and estimation

- The results from exploratory analyses can be misleading
 - If data do not meet assumptions of model or method selected
 - If data have quirks that are idiosyncratic to the sample selected
 - If some cases are extreme relative to others
 - If constraints made by analysis are implausible

- Sometimes, exploratory analyses are needed
 - Must construct an analysis that capitalizes on the known features of data
 - There are better ways to conduct such analyses

- Often, exploratory analyses are not needed
 - But are conducted anyway – see a lot of reports of scale development that start with the idea that a construct has a certain number of dimensions

ADVANCED MATRIX OPERATIONS

- To demonstrate some advanced matrix algebra, we will make use of data
- I collected data SAT test scores for both the Math (SATM) and Verbal (SATV) sections of 1,000 students
- The descriptive statistics of this data set are given below:

Statistic	SATV	SATM
Mean	499.3	498.3
SD	49.8	81.2

Correlation

SATV	1.00	0.78
SATM	0.78	1.00

- For a square matrix Σ with p rows/columns, the trace is the sum of the diagonal elements:

$$tr\Sigma = \sum_{i=1}^p a_{ii}$$

- For our data, the trace of the correlation matrix is 2
 - For all correlation matrices, the trace is equal to the number of variables as all diagonal elements are 1
- The trace is the **total variance** in principal components analysis
 - Used as a target to recover when applying statistical models

- A square matrix can be characterized by a scalar value called a determinant:

$$\det \Sigma = |\Sigma|$$

- Calculation of the determinant by hand is tedious
 - Our determinant was 0.3916
 - Computers can have difficulties with this calculation (unstable in cases)
- The determinant is useful in statistics:
 - Shows up in multivariate statistical distributions
 - Is a measure of “generalized” variance of multiple variables
- If the determinant is positive, the matrix is called **positive definite**
 - Is invertible
- If the determinant is not positive, the matrix is called **non-positive definite**
 - Not invertible

- A square matrix Λ is said to be orthogonal if:

$$\Lambda\Lambda^T = \Lambda^T\Lambda = \mathbf{I}$$

- Orthogonal matrices are characterized by two properties:

1. The dot product of all row vector multiples is the zero vector
 - Meaning vectors are orthogonal (or uncorrelated)
2. For each row vector, the sum of all elements is one
 - Meaning vectors are “normalized”

- The matrix above is also called **orthonormal**

- **The diagonal is equal to 1 (each vector has a unit length)**

- Orthonormal matrices are used in principal components and exploratory factor analysis

- A square matrix Σ can be decomposed into a set of eigenvalues λ and a set of eigenvectors $\mathbf{e}$
$$\Sigma \mathbf{e} = \lambda \mathbf{e}$$
- Each eigenvalue has a corresponding eigenvector
 - The number equal to the number of rows/columns of Σ
 - The eigenvectors are all orthogonal
- Principal components analysis uses eigenvalues and eigenvectors to reconfigure data

- In our SAT example, the two eigenvalues obtained were:

$$\lambda_1 = 1.78$$

$$\lambda_2 = 0.22$$

- The two eigenvectors obtained were:

$$\mathbf{e}_1 = \begin{bmatrix} 0.71 \\ 0.71 \end{bmatrix}; \mathbf{e}_2 = \begin{bmatrix} 0.71 \\ -0.71 \end{bmatrix}$$

- These terms will have much greater meaning principal components analysis

```
#correlation matrix of SAT data
sat_corrmat = cor(data01)

#eigenvalues and eigenvectors of correlation matrix:
sat_eigen = eigen(x = sat_corrmat, symmetric = TRUE)
```

```
> sat_eigen$values
[1] 1.7752238 0.2247762
> sat_eigen$vectors
      [,1]      [,2]
[1,] 0.7071068 -0.7071068
[2,] 0.7071068  0.7071068
```

- Using the eigenvalues and eigenvectors, we can reconstruct the original matrix using a spectral decomposition:

$$\mathbf{\Sigma} = \sum_{i=1}^p \lambda_i \mathbf{e}_i \mathbf{e}_i^T$$

- For our example, we can get back to our original matrix:

$$\mathbf{R}_1 = \lambda_1 \mathbf{e}_1 \mathbf{e}_1^T = 1.78 \begin{bmatrix} .71 \\ .71 \end{bmatrix} \begin{bmatrix} .71 & .71 \end{bmatrix} = \begin{bmatrix} .89 & .89 \\ .89 & .89 \end{bmatrix}$$

$$\mathbf{R}_2 = \mathbf{R}_1 + \lambda_2 \mathbf{e}_2 \mathbf{e}_2^T = \begin{bmatrix} .89 & .89 \\ .89 & .89 \end{bmatrix} + 0.22 \begin{bmatrix} .71 \\ -.71 \end{bmatrix} \begin{bmatrix} .71 & -.71 \end{bmatrix} = \begin{bmatrix} 1.00 & 0.78 \\ 0.78 & 1.00 \end{bmatrix}$$

```
> #demonstration of spectral decomposition
> corr_rank1 = sat_eigen$values[1] * sat_eigen$vectors[,1] %*% t(sat_eigen$vectors[,1])
> corr_rank1
      [,1]      [,2]
[1,] 0.8876119 0.8876119
[2,] 0.8876119 0.8876119
>
> corr_rank2 = corr_rank1 + sat_eigen$values[2] * sat_eigen$vectors[,2] %*% t(sat_eigen$vectors[,2])
> corr_rank2
      [,1]      [,2]
[1,] 1.0000000 0.7752238
[2,] 0.7752238 1.0000000
```

- The matrix trace is the sum of the eigenvalues:

$$tr(\mathbf{\Sigma}) = \sum_{i=1}^p \lambda_i$$

- In our example, the $tr(\mathbf{R}) = 1.78 + .22 = 2$

- The matrix determinant can be found by the product of the eigenvalues

$$|\mathbf{\Sigma}| = \prod_{i=1}^p \lambda_i$$

- In our example $|\mathbf{R}| = 1.78 * .22 = .3916$

AN INTRODUCTION TO PRINCIPAL COMPONENTS ANALYSIS

- Principal Components Analysis (PCA) is a method for re-expressing the covariance (or often correlation) between a set of variables
 - The re-expression comes from creating a set of new variables (linear combinations) from the original variables

- PCA has two objectives:
 1. Data reduction
 - Moving from many original variables down to a few “components”

 2. Interpretation
 - Determining which original variables contribute most to the new “components”

- The goal of PCA is to find a set of k principal components that:
 - Are composite variables (linear combinations of the original variables)
 - Is much smaller in number than the original set of V variables
 - Accounts for nearly all the total variance
 - Total variance = trace of covariance/correlation matrix
- If these two goals can be accomplished, then the set of k principal components contains almost as much information as the original V variables
 - Meaning – the components can now replace the original variables in any subsequent analyses

- PCA analyses proceed by seeking the answers to two questions:
 1. How many components (new variables) are needed to “adequately” represent the original data?
 - The term adequately is fuzzy (and will be in the analysis)
 2. (once #1 has been answered): What does each component represent?
 - The term “represent” is also fuzzy

- PCA often reveals relationships between variables that were not previously suspected
 - New interpretations of data and variables often stem from PCA
- PCA usually serves as more of a means to an end rather than an end in itself
 - Components (the new variables) are often used in other statistical techniques
 - Multiple regression/ANOVA
 - Cluster analysis
- Unfortunately, PCA is often intermixed with exploratory factor analysis (EFA)
 - Many early EFA estimation methods used PCA-like matrix decompositions
 - But...the models are very different
 - PCA → Summative (variables add together to make a component)
 - EFA → Reflective (variables are predicted by the latent traits)
- PCA methods for EFA shouldn't be used
 - Not likelihood-based (worse protection for missing data)
 - Don't. Please don't. Please make it stop.

- Notation: Z are our new components and $\mathbf{Y}$ is our original data matrix (with N observations and V variables)
 - We will let p be our index for a subject

- The new components are linear combinations:

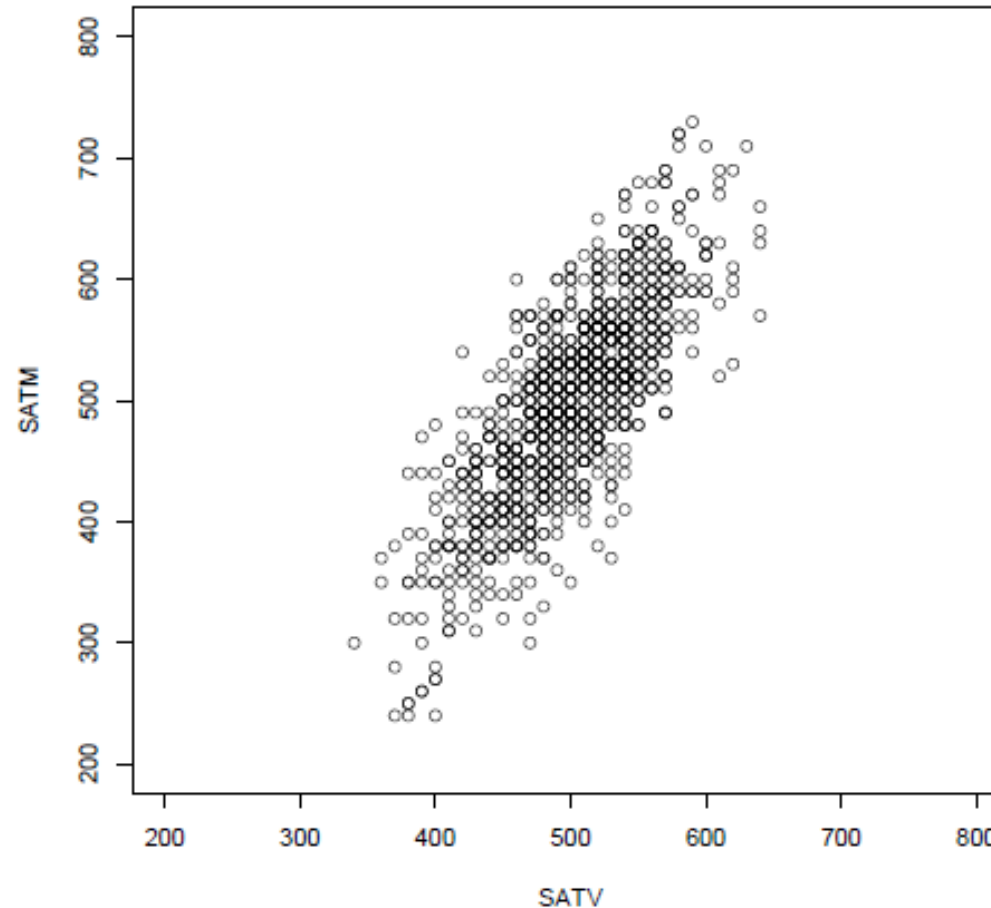
$$\begin{aligned}Z_{p1} &= \mathbf{e}_1^T \mathbf{Y} = e_{11}Y_{p1} + e_{21}Y_{p2} + \cdots + e_{V1}Y_{pV} \\Z_{p2} &= \mathbf{e}_2^T \mathbf{Y} = e_{12}Y_{p1} + e_{22}Y_{p2} + \cdots + e_{V2}Y_{pV} \\&\vdots \\Z_{pV} &= \mathbf{e}_V^T \mathbf{Y} = e_{1V}Y_{p1} + e_{2V}Y_{p2} + \cdots + e_{VV}Y_{pV}\end{aligned}$$

- The weights of the components (e_{jk}) come from the eigenvectors of the covariance or correlation matrix for component k and variable j

- The components (Z) are formed by the weights of the eigenvectors of the covariance or correlation matrix of the original data
 - The variance of a component is given by the eigenvalue associated with the eigenvector for the component
- Using the eigenvalue and eigenvectors means:
 - Each successive component has lower variance
 - $\text{Var}(Z_1) > \text{Var}(Z_2) > \dots > \text{Var}(Z_v)$
 - All components are uncorrelated
 - The sum of the variances of the principal components is equal to the total variance:

$$\sum_{v=1}^V \text{Var}(Z_v) = \text{tr}\mathbf{\Sigma} = \sum_{v=1}^V \lambda_v$$

- We will now conduct a PCA on the **correlation matrix** of our sample data
 - This example is given for demonstration purposes – typically we will not do PCA on small numbers of variables



- The R function that does principal components is called `prcomp()`

```
#PCA of Correlation matrix (scale. = TRUE)
sat_pca_corr = prcomp(x = data01, scale. = TRUE)
```

```
#show the results (compare to eigenvalues/eigenvectors)
sat_pca_corr
```

```
#show the summary statistics
summary(sat_pca_corr)
```

```
> sat_pca_corr
```

Standard deviations:

```
[1] 1.3323753 0.4741057
```

Rotation:

	PC1	PC2
SATV	-0.7071068	0.7071068
SATM	-0.7071068	-0.7071068

```
>
```

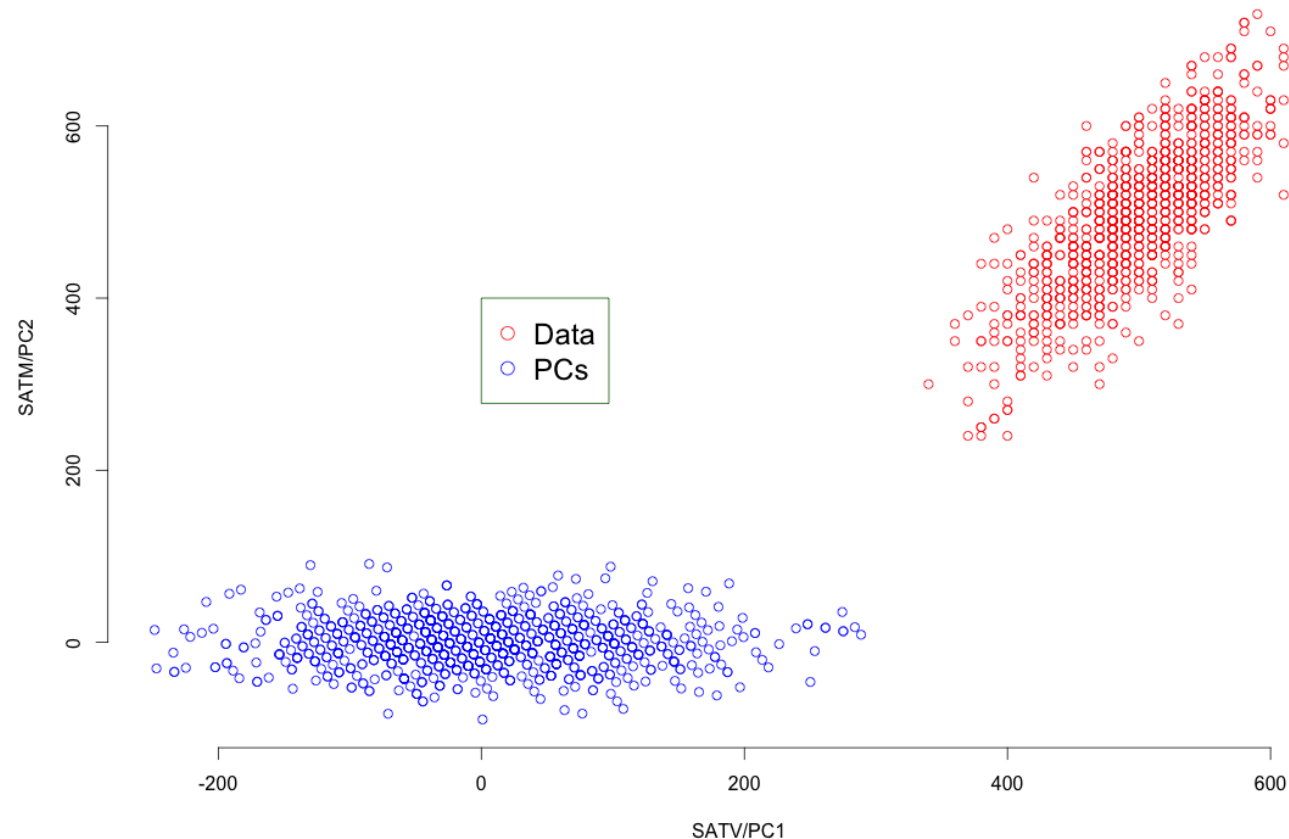
```
> #show the summary statistics
```

```
> summary(sat_pca_corr)
```

Importance of components:

	PC1	PC2
Standard deviation	1.3324	0.4741
Proportion of Variance	0.8876	0.1124
Cumulative Proportion	0.8876	1.0000

- Plotting the components and the original data side by side reveals the nature of PCA:
 - Shown from PCA of covariance matrix



- In past 25 years:
 - An exponential increase in the accessibility of gambling
 - An increased rate of with problem or pathological gambling (Volberg, 2002, Welte et al., 2009)
- Hence, there is a need to better:
 - Understand the underlying causes of the disorder
 - Reliably identify potential pathological gamblers
 - Provide effective treatment interventions



➤ To be diagnosed as a pathological gambler, an individual must meet 5 of 10 defined criteria:

1. Is preoccupied with gambling
2. Needs to gamble with increasing amounts of money in order to achieve the desired excitement
3. Has repeated unsuccessful efforts to control, cut back, or stop gambling
4. Is restless or irritable when attempting to cut down or stop gambling
5. Gambles as a way of escaping from problems or relieving a dysphoric mood
6. After losing money gambling, often returns another day to get even
7. Lies to family members, therapist, or others to conceal the extent of involvement with gambling
8. Has committed illegal acts such as forgery, fraud, theft, or embezzlement to finance gambling
9. Has jeopardized or lost a significant relationship, job, educational, or career opportunity because of gambling
10. Relies on others to provide money to relieve a desperate financial situation caused by gambling

- In order to study the etiology of pathological gambling, more variability in responses was needed
- The Gambling Research Instrument (Feasel, Henson, & Jones, 2002) was created with 41 Likert-type items
 - Items were developed to measure each criterion
- Example items (ratings: Strongly Disagree to Strongly Agree):
 - I worry that I am spending too much money on gambling (C3)
 - There are few things I would rather do than gamble (C1)
- The instrument was used on a sample of experienced gamblers from a riverboat casino in a Flat Midwestern State
 - Casino patrons were solicited after playing roulette

- The GRI used a 6-point Likert scale
 - 1: Strongly Disagree
 - 2: Disagree
 - 3: Slightly Disagree
 - 4: Slightly Agree
 - 5: Agree
 - 6: Strongly Agree

- To meet the assumptions of factor analysis, we will treat these responses as being continuous
 - This is tenuous at best, but often is the case in factor analysis
 - Categorical items would be better....but you'd need another course for how to do that
 - Hint: Item Response Models

- Data were collected from two sources:
 - 112 “experienced” gamblers
 - Many from an actual casino
 - 1192 college students from a “rectangular” midwestern state
 - Many never gambled before
- Today, we will combine both samples and treat them as homogenous – one sample of 1304 subjects

Item	Criterion	Question
GRI1	3	I would like to cut back on my gambling.
GRI3	6	If I lost a lot of money gambling one day, I would be more likely to want to play again the following day.
GRI5	2	I find it necessary to gamble with larger amounts of money (than when I first gambled) for gambling to be exciting.
GRI9	4	I feel restless when I try to cut down or stop gambling.
GRI10	1	It bothers me when I have no money to gamble.
GRI13	3	I find it difficult to stop gambling.
GRI14	2	I am drawn more by the thrill of gambling than by the money I could win.
GRI18	9	My family, coworkers, or others who are close to me disapprove of my gambling.
GRI21	1	It is hard to get my mind off gambling.
GRI23	5	I gamble to improve my mood.

- To show how PCA works with a larger set of items, we will examine the 10 GRI items (the ones that fit a one-factor CFA model)

- **TO DO THIS YOU MUST IMAGINE:**
 - THESE WERE THE ONLY 10 ITEMS YOU HAD
 - YOU WANTED TO REDUCE THE 10 ITEMS INTO 1 OR 2 COMPONENT VARIABLES

- CAPITAL LETTERS ARE USED AS YOU SHOULD NEVER DO A PCA AFTER RUNNING A CFA – THEY ARE FOR DIFFERENT PURPOSES!

Question #1: How Many Components?

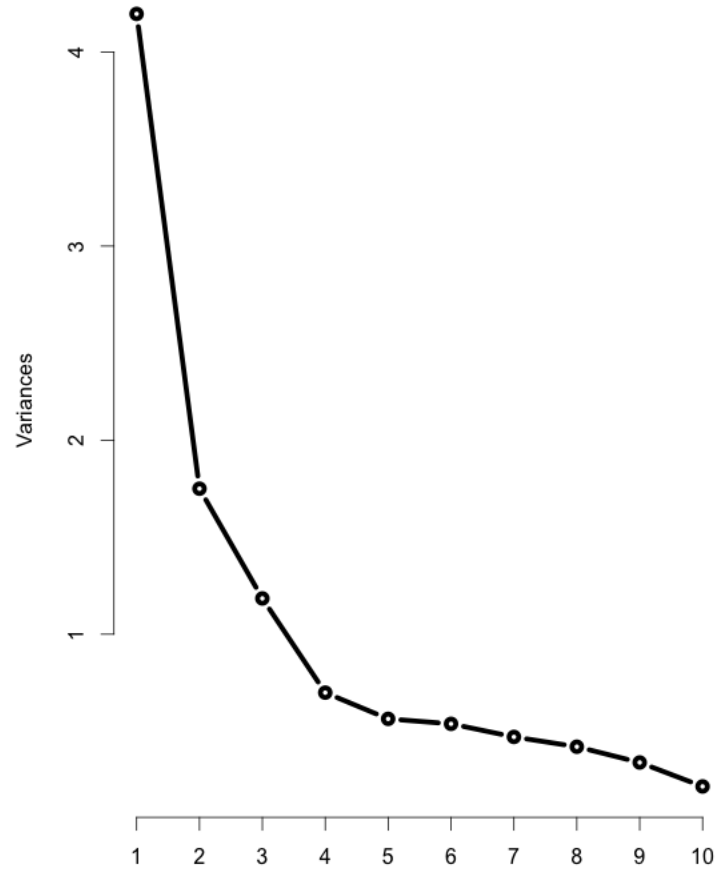
- To answer the question of how many components, two methods are used:
 - Scree plot of eigenvalues (looking for the “elbow”)
 - Variance accounted for (should be > 70%)
- We will go with 4 components: (variance accounted for VAC = 75%)
- Variance accounted for is for the **total sample variance**

```
> summary(gambling_pca_cov)
```

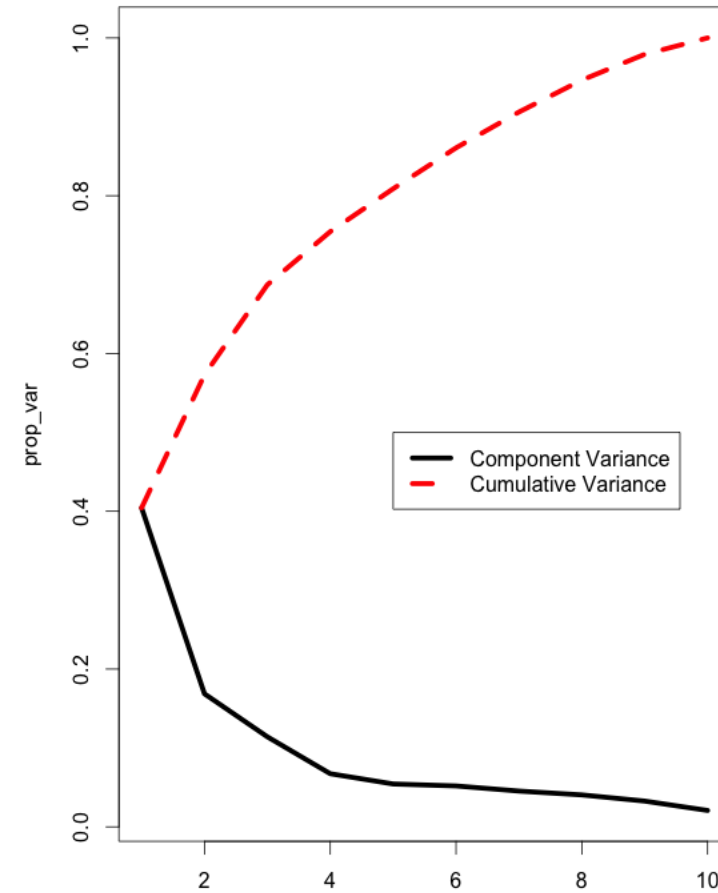
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
Standard deviation	2.0485	1.3229	1.0883	0.83608	0.75096	0.73365	0.68616	0.64836	0.58230	0.46438
Proportion of Variance	0.4043	0.1686	0.1141	0.06736	0.05434	0.05186	0.04537	0.04051	0.03267	0.02078
Cumulative Proportion	0.4043	0.5730	0.6871	0.75447	0.80881	0.86067	0.90604	0.94655	0.97922	1.00000

Scree Plot of PCA Eigenvalues



Proportion of Variance Explained by Component



- To answer question #2 – we look at the weights of the eigenvectors
 - This is the unrotated solution

Rotation:

	PC1	PC2	PC3	PC4
X1	-0.3126472	0.10459331	0.1662869	-0.843735151
X3	-0.2262923	0.07828499	0.1775587	-0.133596441
X5	-0.3199396	0.09403404	0.2294114	0.025555994
X9	-0.2433589	0.08687880	0.1596160	0.030715826
X10	-0.2798829	0.08714575	0.2352173	0.084941435
X13	-0.3293561	0.15569572	0.1863762	0.220349288
X14	-0.4493615	-0.86166945	-0.2311989	0.005504759
X18	-0.3471876	0.40458523	-0.8343757	-0.028338155
X21	-0.2720582	0.13735301	0.0617232	0.198445605
X23	-0.3258344	0.09836941	0.1385856	0.415552100

- Using the weights of the eigenvectors, we can create four new variables – the four principal components

	PC1	PC2	PC3	PC4	PC
1	-2.54488887	1.28795898	1.58233323	-1.82303154	-1.
2	2.18514082	0.36461255	0.19847109	0.26427683	0.
3	-0.61790497	0.66444313	0.24031780	0.37842837	0.
4	0.14430776	1.12694087	1.53931389	-0.13114293	-0.
5	0.14430776	1.12694087	1.53931389	-0.13114293	-0.
6	2.18514082	0.36461255	0.19847109	0.26427683	0.
7	2.18514082	0.36461255	0.19847109	0.26427683	0.
8	2.18514082	0.36461255	0.19847109	0.26427683	0.
9	1.87249362	0.46920586	0.36475803	-0.57945832	0.
10	1.87249362	0.46920586	0.36475803	-0.57945832	0.
11	2.18514082	0.36461255	0.19847109	0.26427683	0.
12	2.18514082	0.36461255	0.19847109	0.26427683	0.
13	2.18514082	0.36461255	0.19847109	0.26427683	0.
14	2.18514082	0.36461255	0.19847109	0.26427683	0.

- Each of these is uncorrelated with each other
 - The variance of each is equal to the corresponding eigenvalue
- We would then use these in subsequent analyses

- PCA is a **data reduction** technique that relies on the mathematical properties of eigenvalues and eigenvectors
 - **Used to create new variables (small number) out of the old data (lots of variables)**
 - The new variables are principal components (they are not factor scores)
- PCA appeared first in the psychometric literature
 - Many “factor analysis” methods used variants of PCA before likelihood-based statistics were available
- Currently, PCA (or variants) methods are the default option in SPSS and SAS (PROC FACTOR)

- The typical PCA analysis also has a few statistical concerns
 - Some of these can be solved if you know what you are doing
 - The typical analysis (using program defaults) does not solve these
- Missing data is omitted using listwise deletion – biases possible
 - Could use ML to estimate covariance matrix, but then would have to assume multivariate normality
 - Could use MI to impute data
- The distributions of variables can be anything...but variables with much larger variances will look like they contribute more to each component
 - Could standardize variables – but some can't be standardized easily (think gender)
- The lack of standard errors makes the component weights (eigenvector elements) hard to interpret
 - Can use a resampling/bootstrap analysis to get SEs (but not easy to do)

- My issues with PCA involve the two questions in need of answers for any use of PCA:
 1. The number of components needed is not based on a statistical hypothesis test and hence is subjective
 - Variance accounted for is a descriptive measure
 - No statistical test for whether an additional component significantly accounts for more variance
 2. The relative meaning of each component is questionable at best and hence is subjective
 - Typical packages provide no standard errors for each eigenvector weight (can be obtained in bootstrap analyses)
 - No definitive answer for component composition
- In sum, I feel it is very easy to be misled (or purposefully mislead) with PCA

EXPLORATORY FACTOR ANALYSIS

- **EFA:** “Determine nature and number of latent variables that account for observed variation and covariation among set of observed indicators ($\approx$ items or variables)”
 - In other words, what causes these observed responses?
 - Summarize patterns of correlation among indicators
 - Solution is an end (i.e., is of interest) in and of itself

- Compared with **PCA:** “Reduce multiple observed variables into fewer components that summarize their variance”
 - In other words, how can I abbreviate this set of variables?
 - Solution is usually a means to an end

- You will see many different types of methods for “extraction” of factors in EFA
 - Many are PCA-based
 - Most were developed before computers became relevant or likelihood theory was developed
- You can ignore all of them and focus on one:

Only Use Maximum Likelihood for EFA

- The maximum likelihood method of EFA extraction:
 - Uses the same log-likelihood as confirmatory factor analyses/SEM
 - Default assumption: multivariate normal distribution of data
 - Provides consistent estimates with good statistical properties (assuming you have a large enough sample)
 - Missing data using all the data that was observed (MAR)
 - Is consistent with modern statistical practices

- EFAs proceed by seeking the answers to two questions:
(the same questions posed in PCA; but with different terms)
- 1. How many latent factors are needed to “adequately” represent the original data?
 - “Adequately” = does a given EFA model fit well?
- 2. (once #1 has been answered): What does each factor represent?
 - The term “represent” is fuzzy

- Factor analysis works by hypothesizing that a set of latent factors helps to determine a person's response to a set of variables

- This can be explained by a system of simultaneous linear models

- Here Y = observed data, p = person, v = variable, F = factor score (Q factors)

$$Y_{p1} = \mu_{y_1} + \lambda_{11}F_{p1} + \lambda_{12}F_{p2} + \cdots + \lambda_{1Q}F_{pQ} + e_{p1}$$

$$Y_{p2} = \mu_{y_2} + \lambda_{21}F_{p1} + \lambda_{22}F_{p2} + \cdots + \lambda_{2Q}F_{pQ} + e_{p2}$$

$$\vdots$$

$$Y_{pV} = \mu_{y_V} + \lambda_{V1}F_{p1} + \lambda_{V2}F_{p2} + \cdots + \lambda_{VQ}F_{pQ} + e_{pV}$$

- μ_{y_v} = mean for variable v
- λ_{vf} = factor loading for variable v onto factor f (regression slope)
 - Factors are assumed distributed MVN with zero mean and (for EFA) identity covariance matrix (uncorrelated factors – to start)
- e_{pv} = residual for person p and variable v
 - Residuals are assumed distributed MVN (across items) with a zero mean and a diagonal covariance matrix Ψ containing the unique variances
- Often, this gets shortened into matrix form:

$$\mathbf{Y}_p = \boldsymbol{\mu}_Y + \boldsymbol{\Lambda}\mathbf{F}_p^T + \mathbf{e}_p$$

- Maximum likelihood EFA assumes the data follow a multivariate normal distribution
 - The basis for the log-likelihood function (same log-likelihood we have used in every analysis to this point)
- The log-likelihood function depends on two sets of parameters: the mean vector and the covariance matrix
 - Mean vector is saturated (just uses the item means for item intercepts) – so it is often not thought of in analysis
 - Denoted as $\mu_Y = \mu_I$
 - Covariance matrix is what gives “factor structure”
 - EFA models provide a structure for the covariance matrix

- The covariance matrix is modeled based on how it would look if a set of hypothetical (latent) factors had caused the data
- For an analysis measuring F factors, each item in the EFA:
 - Has 1 unique variance parameter
 - Has F factor loadings
- The initial estimation of factor loadings is conducted based on the assumption of uncorrelated factors
 - Assumption is dubious at best – yet is the cornerstone of the analysis

➤ The factor model implied covariance matrix is $\Sigma_Y = \Lambda\Phi\Lambda^T + \Psi$

➤ Where:

- Σ_Y = model implied covariance matrix of the observed data (size $I \times I$)
- Λ = matrix of factor loadings (size $I \times F$)
 - In EFA: all terms in Λ are estimated
- Φ = factor covariance matrix (size $F \times F$)
 - In EFA: $\Phi = \mathbf{I}$ (all factors have variances of 1 and covariances of 0)
 - In CFA: this is estimated
- Ψ = matrix of unique (residual) variances (size $I \times I$)
 - In EFA: Ψ is diagonal by default (no residual covariances)

➤ Therefore, the EFA model-implied covariance matrix is:

$$\Sigma_Y = \Lambda\Lambda^T + \Psi$$

- Under the ML method for EFA, the same rules of identification apply to EFA as to Path Analysis
 - T-rule: Total number of EFA model parameters must not exceed unique elements in saturated covariance matrix of data
 - For an analysis with a number of factors F and a set number of items I there are $F*I + I = I(F + 1)$ EFA model parameters
 - As we will see, there must be $\frac{F(F-1)}{2}$ constraints for the model to work
 - Therefore, $I(F + 1) - \frac{F(F-1)}{2} < \frac{I(I+1)}{2}$
 - Local-identification: each portion of the model must be locally identified
 - With all factor loadings estimated local identification fails
 - No way of differentiating factors without constraints

- The EFA model typically imposes the following constraint (Lawley & Maxwell, 1962):

$$\Lambda^T \Psi \Lambda = \Delta$$

such that Δ is a diagonal matrix

- This puts $\frac{F(F-1)}{2}$ constraints on the model (that many fewer parameters to estimate)
- This constraint is not well known – and how it functions is hard to describe
 - For a 1-factor model, the results of EFA and CFA will match
- Note: the other methods of EFA “extraction” avoid this constraint by not being statistical models in the first place
 - PCA-based routines rely on matrix properties to resolve identification

- The EFA constraints provide some detailed assumptions about the nature of the factor model and how it pertains to the data

- For example, take a 2-factor model (one constraint):

$$\sum_{v=1}^V \psi_v^2 \prod_{f=1}^{Q=2} \lambda_{vf} = 0$$

- In short, some combinations of factor loadings and unique variances (across and within items) cannot happen
 - This goes against most of our statistical constraints – which must be justifiable and understandable (therefore testable)
 - This constraint is not testable in CFA

- Given the model parameters, the EFA model is estimated by maximizing the multivariate normal log-likelihood
 - For the data

$$\log L = \log \left[(2\pi)^{-\frac{NV}{2}} |\mathbf{\Sigma}|^{-\frac{N}{2}} \exp \left[\sum_{p=1}^N -\frac{(\mathbf{Y}_p - \boldsymbol{\mu}_y)^T \mathbf{\Sigma}^{-1} (\mathbf{Y}_p - \boldsymbol{\mu}_y)}{2} \right] \right] =$$
$$-\frac{NV}{2} \log(2\pi) - \frac{N}{2} \log(|\mathbf{\Sigma}|) - \sum_{p=1}^N \frac{(\mathbf{Y}_p - \boldsymbol{\mu}_y)^T \mathbf{\Sigma}^{-1} (\mathbf{Y}_p - \boldsymbol{\mu}_y)}{2}$$

- Under EFA, this becomes:

$$\log L = -\frac{NV}{2} \log(2\pi) - \frac{N}{2} \log(|\mathbf{\Lambda}\mathbf{\Lambda}^T + \mathbf{\Psi}|) - \sum_{p=1}^N \frac{(\mathbf{Y}_p - \boldsymbol{\mu}_I)^T (\mathbf{\Lambda}\mathbf{\Lambda}^T + \mathbf{\Psi})^{-1} (\mathbf{Y}_p - \boldsymbol{\mu}_I)}{2}$$

- The parameters of the EFA model under ML retain the same benefits and consequences of any model (i.e., CFA)
 - Asymptotically (large N) they are consistent, normal, and efficient
 - Missing data are “skipped” in the likelihood, allowing for incomplete observations to contribute (assumed MAR)
- Furthermore, the same types of model fit indices are available in EFA as are in CFA
- As with CFA, though, an EFA model must be a close approximation to the saturated model covariance matrix if the parameters are to be believed
 - This is a marked difference between EFA in ML and EFA with other methods – quality of fit is statistically rigorous

ML EFA WITH BASE R FUNCTION FACTANAL (THE BAD WAY)

- The base R program has the factanal() function that conducts ML-based EFA
 - But it is very limited
- Although the function use ML, you still cannot have missing data in the analysis
 - BAD R!
- We will remove cases with any missing data (listwise deletion) and proceed
- We will also not use a rotation method at first as to show how default constraints in EFA with ML are ridiculous

```
#principal components analysis of sampling data
#listwise removal of missing data (common in PCA -- but still a problem)
data02a = data02[which(is.na(data02$X1)==FALSE & is.na(data02$X3)==FALSE & is.na(data02$X5)==FALSE & is.na(data02$X9)==FALSE & is.na(data02$X10)==FALSE &
is.na(data02$X13)==FALSE & is.na(data02$X14)==FALSE & is.na(data02$X18)==FALSE & is.na(data02$X21)==FALSE & is.na(data02$X23)==FALSE),]
```

Step #1: Determine Number of Factors

- The EFA `factanal()` function provides a rudimentary test for model fit
- Remember the saturated model from path analysis?
 - All covariances estimated
- The model fit tests the solution from EFA vs the saturated model
 - EFA 1-factor model shown
- The goal is to find a model that fits well

```
> EFA_1factor = factanal(x = data02a, factors = 1, rotation = "none")  
> EFA_1factor
```

```
Call:  
factanal(x = data02a, factors = 1, rotation = "none")
```

```
Uniquenesses:  
      X1      X3      X5      X9     X10     X13     X14     X18     X21     X23  
0.677 0.728 0.550 0.417 0.527 0.488 0.857 0.816 0.538 0.573
```

```
Loadings:  
      Factor1  
X1  0.569  
X3  0.521  
X5  0.670  
X9  0.764  
X10 0.688  
X13 0.715  
X14 0.378  
X18 0.429  
X21 0.680  
X23 0.653
```

```
      Factor1  
SS loadings      3.828  
Proportion Var   0.383
```

Test of the hypothesis that 1 factor is sufficient.
The chi square statistic is 161.14 on 35 degrees of freedom.
The p-value is 4.23e-18

➤ One factor:

Test of the hypothesis that 1 factor is sufficient.
The chi square statistic is 161.14 on 35 degrees of freedom.
The p-value is 4.23e-18

➤ Two factors:

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 56.43 on 26 degrees of freedom.
The p-value is 0.000497

➤ Three factors:

Test of the hypothesis that 3 factors are sufficient.
The chi square statistic is 31.24 on 18 degrees of freedom.
The p-value is 0.027

➤ Four factors:

Test of the hypothesis that 4 factors are sufficient.
The chi square statistic is 10.2 on 11 degrees of freedom.
The p-value is 0.512

- The EFA model imposes the following constraint:

$$\Lambda^T \Psi \Lambda = \Delta$$

such that Δ is a diagonal matrix

- Here are the Δ matrices from each analysis:

```
> #constraint demonstration (lambda^T psi-1 lambda = diag)
> Lambda = matrix(EFA_2factor$loadings, ncol=2)
> Psi = diag(EFA_2factor$uniquenesses)
> t(Lambda) %*% solve(Psi) %*% Lambda
```

```
      [,1]      [,2]
[1,] 7.698164e+00 -1.94289e-16
[2,] 4.163336e-17  5.57827e-01
```

```
> #constraint demonstration (lambda^T psi-1 lambda = diag)
> Lambda = matrix(EFA_3factor$loadings, ncol=3)
> Psi = diag(EFA_3factor$uniquenesses)
> t(Lambda) %*% solve(Psi) %*% Lambda
```

```
      [,1]      [,2]      [,3]
[1,] 2.026141e+02  2.220446e-16 -3.122502e-16
[2,] -2.220446e-16  2.683701e+00 -1.873501e-16
[3,] -2.636780e-16 -1.942890e-16  4.352294e-01
```

```
> Lambda = matrix(EFA_4factor$loadings, ncol=4)
> Psi = diag(EFA_4factor$uniquenesses)
> t(Lambda) %*% solve(Psi) %*% Lambda
```

```
      [,1]      [,2]      [,3]      [,4]
[1,] 2.023692e+02  0.000000e+00 -2.081668e-16  1.318390e-16
[2,] 0.000000e+00  4.558736e+00  7.524363e-17 -1.139496e-16
[3,] -1.942890e-16  1.860491e-16  9.039206e-01 -2.110721e-16
[4,] 6.245005e-17 -1.070108e-16 -2.041332e-16  3.539411e-01
```

Step #2: Interpreting the Best Model

- As the four-factor solution fit best, we will interpret it
- Unrotated solution of factor loadings:

Loadings:				
	Factor1	Factor2	Factor3	Factor4
X1	0.356	0.457	0.127	0.346
X3	0.322	0.403		
X5	0.452	0.480	0.174	
X9	0.468	0.630	0.196	-0.144
X10	0.458	0.502	0.186	
X13	0.509	0.494		
X14	0.292	0.227		
X18	0.316	0.294	-0.141	0.128
X21	0.491	0.548	-0.408	
X23	0.997			

What???

FACTOR LOADING ROTATIONS IN EFA

- Transformations of the factor loadings are possible as the matrix of factor loadings is only unique up to an orthogonal transformation
 - Don't like the solution? Rotate!
- Historically, rotations use the properties of matrix algebra to adjust the factor loadings to more interpretable numbers
- Modern versions of rotations/transformations rely on “target functions” that specify what a “good” solution should look like
 - The details of the modern approach are lacking in most texts

- Multiple types of rotations exist but two broad categories seem to dominate how they are discussed:
- **Orthogonal rotations:** rotations that force the factor correlation to zero (orthogonal factors). The name orthogonal relates to the angle between axes of factor solutions being 90 degrees. The most prevalent is the varimax rotation.
- **Oblique rotations:** rotations that allow for non-zero factor correlations. The name orthogonal relates to the angle between axes of factor solutions not being 90 degrees. The most prevalent is the promax rotation.
 - These rotations provide an estimate of “factor correlation”

- Classical orthogonal rotation algorithms work by defining a new rotated set of factor loadings Λ^* as a function of the original (non-rotated) loadings Λ and an orthogonal rotation matrix $\mathbf{T}$

$$\Lambda^* = \Lambda \mathbf{T}$$

where: $\mathbf{T}\mathbf{T}^T = \mathbf{T}^T\mathbf{T} = \mathbf{I}$

- These rotations do not alter the fit of the model as

$$\Sigma_Y = \Lambda^* \Lambda^{*T} + \Psi = \Lambda \mathbf{T} (\Lambda \mathbf{T})^T + \Psi = \Lambda \mathbf{T} \mathbf{T}^T \Lambda^T + \Psi = \Lambda \Lambda^T + \Psi$$

- Most studies using EFA use the classical rotation mechanisms, likely due to insufficient training
- Modern methods for rotations rely on the use of a target function for how an optimal loading solution should look

$$\mathbf{L} = \begin{bmatrix} x & 0 & 0 \\ x & 0 & 0 \\ x & 0 & 0 \\ x & 0 & 0 \\ 0 & x & 0 \\ 0 & x & 0 \\ 0 & 0 & x \\ 0 & 0 & x \\ 0 & 0 & x \end{bmatrix}$$

From Browne (2001)

- Given a target function, rotation algorithms seek to find a rotated solution that simultaneously:
 1. Minimizes the distance between the rotated solution and the original factor loadings
 2. Fits best to the target function
- Rotation algorithms are typically iterative – meaning they can fail to converge
- Rotation searches typically have multiple optimal values
 - Need many restarts

- The Varimax rotation brought about the following loadings

Loadings:				
	Factor1	Factor2	Factor3	Factor4
X1	0.301	0.209	0.114	0.569
X3	0.360	0.213	0.113	0.307
X5	0.532	0.224	0.202	0.301
X9	0.714	0.291	0.157	0.239
X10	0.597	0.231	0.202	0.234
X13	0.414	0.461	0.236	0.268
X14	0.224	0.129	0.162	0.242
X18	0.147	0.346	0.144	0.246
X21	0.300	0.746	0.184	0.166
X23	0.266	0.270	0.904	0.188

- Are these better for interpretation?
- Also note: no factor correlation

- The Promax rotation brought about the following loadings:

Loadings:				
	Factor1	Factor2	Factor3	Factor4
X1			-0.104	0.861
X3	0.249			0.308
X5	0.530			0.171
X9	0.885			
X10	0.699			
X13	0.254		0.368	0.101
X14				0.274
X18	-0.126		0.314	0.269
X21			0.906	-0.106
X23		1.037		

- It also brought about the following factor correlations:

Factor Correlations:				
	Factor1	Factor2	Factor3	Factor4
Factor1	1.000	0.635	-0.628	-0.631
Factor2	0.635	1.000	-0.736	-0.852
Factor3	-0.628	-0.736	1.000	0.768
Factor4	-0.631	-0.852	0.768	1.000

EFA VIA CFA

- We can conduct exploratory analysis using a CFA model
 - Need to set the right number of constraints for identification
 - We set the value of factor loadings for a few items on a few of the factors
 - Typically to zero (my usual thought)
 - Sometimes to one (Brown, 2002)
 - We keep the factor covariance matrix as an identity
 - Uncorrelated factors (as in EFA) with variances of one

- Benefits of using CFA for exploratory analyses:
 - CFA constraints remove rotational indeterminacy of factor loadings – no rotating is needed (or possible)
 - Defines factors with potentially less ambiguity
 - Constraints are easy to see
 - For some software (SAS and SPSS), we get much more model fit information

- To do EFA with CFA, you must:
 - Fix factor loadings (set to either zero or one)
 - Use “row echelon” form :
 - One item has only one factor loading estimated
 - One item has only two factor loadings estimated
 - One item has only three factor loadings estimated
 - Fix factor covariances
 - Set all to 0
 - Fix factor variances
 - Set all to 1

- We can use lavaan to do CFA...here is the syntax for the one factor model
 - The `~` is the key → Factor name to the left, items measuring it to the right

```
#one factor CFA
CFA_1factor.syntax = "
factor1 =~ X1 + X3 + X5 + X9 + X10 + X13 + X14 + X18 + X21 + X23
"

#for comparison with EFA we are using standardized factors (var = 1; mean = 0)
CFA_1factor.model = cfa(model = CFA_1factor.syntax, data = data02a, estimator = "MLR", std.lv = TRUE)
summary(CFA_1factor.model, fit.measures = TRUE, standardized = TRUE)
```

```
> summary(CFA_1factor.model, fit.measures = TRUE, standardized = TRUE)
lavaan 0.6-19 ended normally after 21 iterations
```

Estimator	ML
Optimization method	NLMINB
Number of model parameters	20
Number of observations	1333

Model Test User Model:

	Standard	Scaled
Test Statistic	161.846	104.001
Degrees of freedom	35	35
P-value (Chi-square)	0.000	0.000
Scaling correction factor		1.556
Yuan-Bentler correction (Mplus variant)		

Model Test Baseline Model:

Test statistic	4148.081	2238.585
Degrees of freedom	45	45
P-value	0.000	0.000
Scaling correction factor		1.853

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.969	0.969
Tucker-Lewis Index (TLI)	0.960	0.960
Robust Comparative Fit Index (CFI)		0.974
Robust Tucker-Lewis Index (TLI)		0.966

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-16575.733	-16575.733
Scaling correction factor for the MLR correction		3.031
Loglikelihood unrestricted model (H1)	-16494.810	-16494.810
Scaling correction factor for the MLR correction		2.092
Akaike (AIC)	33191.466	33191.466
Bayesian (BIC)	33295.370	33295.370
Sample-size adjusted Bayesian (SABIC)	33231.839	33231.839

Root Mean Square Error of Approximation:

RMSEA	0.052	0.038
90 Percent confidence interval - lower	0.044	0.032
90 Percent confidence interval - upper	0.060	0.045
P-value H_0: RMSEA <= 0.050	0.318	0.997
P-value H_0: RMSEA >= 0.080	0.000	0.000

Robust RMSEA	0.048
90 Percent confidence interval - lower	0.037
90 Percent confidence interval - upper	0.059
P-value H_0: Robust RMSEA <= 0.050	0.604
P-value H_0: Robust RMSEA >= 0.080	0.000

Standardized Root Mean Square Residual:

SRMR	0.028	0.028
------	-------	-------

Parameter Estimates:

Standard errors	Sandwich
Information bread	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
factor1 =~						
X1	0.581	0.039	15.002	0.000	0.581	0.569
X3	0.451	0.038	11.896	0.000	0.451	0.521
X5	0.647	0.041	15.684	0.000	0.647	0.670
X9	0.542	0.034	16.182	0.000	0.542	0.764
X10	0.598	0.034	17.355	0.000	0.598	0.688
X13	0.684	0.037	18.334	0.000	0.684	0.715
X14	0.562	0.038	14.601	0.000	0.562	0.378
X18	0.547	0.038	14.438	0.000	0.547	0.429
X21	0.564	0.036	15.774	0.000	0.564	0.680
X23	0.635	0.033	19.346	0.000	0.635	0.653

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.X1	0.706	0.063	11.130	0.000	0.706	0.677
.X3	0.546	0.043	12.732	0.000	0.546	0.728
.X5	0.513	0.047	10.888	0.000	0.513	0.550
.X9	0.210	0.016	12.932	0.000	0.210	0.417
.X10	0.399	0.043	9.297	0.000	0.399	0.527
.X13	0.447	0.047	9.541	0.000	0.447	0.488
.X14	1.894	0.078	24.226	0.000	1.894	0.857
.X18	1.326	0.096	13.842	0.000	1.326	0.816
.X21	0.370	0.036	10.303	0.000	0.370	0.538
.X23	0.542	0.047	11.570	0.000	0.542	0.573
factor1	1.000				1.000	1.000

Loadings:

Factor1

X1 0.569

X3 0.521

X5 0.670

X9 0.764

X10 0.688

X13 0.715

X14 0.378

X18 0.429

X21 0.680

X23 0.653

Factor1

SS loadings 3.828

Proportion Var 0.383

Test of the hypothesis that 1 factor is sufficient.

The chi square statistic is 161.14 on 35 degrees of freedom.

The p-value is 4.23e-18

- Because our one-factor model fit well, we can stop!
- CFA has more indices of model fit – which can make finding an appropriate solution easier
- CFA also gives you the standard errors for each factor loading, leading to a Wald test to see if it is non-zero
 - No need to use arbitrary .3 cutoff
 - Small note: Although most EFA routines (like factanal) don't give SEs they are certainly attainable under ML theory
- Although we should stop here...We'll continue with the two- and three-factor versions to compare with EFA

```
#two factor CFA: one item removed from factor 2 and zero covariance between factors

CFA_2factor.syntax = "
factor1 =~ X1 + X3 + X5 + X9 + X10 + X13 + X14 + X18 + X21 + X23
factor2 =~      X3 + X5 + X9 + X10 + X13 + X14 + X18 + X21 + X23

factor1 ~ 0*factor2
"

#for comparison with EFA we are using standardized factors (var = 1; mean = 0)
CFA_2factor.model = cfa(model = CFA_2factor.syntax, data = data02a, estimator = "MLR", std.lv = TRUE)
summary(CFA_2factor.model, fit.measures = TRUE, standardized = TRUE)
```


CFA Two-Factor Results

```
> summary(CFA_2factor.model, fit.measures = TRUE, standardized = TRUE)
lavaan 0.6-19 ended normally after 42 iterations
```

Estimator	ML	
Optimization method	NLMINB	
Number of model parameters	29	
Number of observations	1333	
Model Test User Model:		
Test Statistic	Standard	Scaled
Degrees of freedom	56.704	39.189
P-value (Chi-square)	26	26
Scaling correction factor	0.000	0.047
Yuan-Bentler correction (Mplus variant)		1.447
Model Test Baseline Model:		
Test statistic	4148.081	2238.585
Degrees of freedom	45	45
P-value	0.000	0.000
Scaling correction factor		1.853
User Model versus Baseline Model:		
Comparative Fit Index (CFI)	0.993	0.994
Tucker-Lewis Index (TLI)	0.987	0.990
Robust Comparative Fit Index (CFI)		0.995
Robust Tucker-Lewis Index (TLI)		0.992
Loglikelihood and Information Criteria:		
Loglikelihood user model (H0)	-16523.162	-16523.162
Scaling correction factor for the MLR correction		2.671
Loglikelihood unrestricted model (H1)	-16494.810	-16494.810
Scaling correction factor for the MLR correction		2.092
Akaike (AIC)	33104.324	33104.324
Bayesian (BIC)	33254.985	33254.985
Sample-size adjusted Bayesian (SABIC)	33162.865	33162.865

Root Mean Square Error of Approximation:

RMSEA	0.030	0.020
90 Percent confidence interval - lower	0.019	0.007
90 Percent confidence interval - upper	0.040	0.029
P-value H_0: RMSEA <= 0.050	0.999	1.000
P-value H_0: RMSEA >= 0.080	0.000	0.000

Robust RMSEA	0.023
90 Percent confidence interval - lower	0.003
90 Percent confidence interval - upper	0.038
P-value H_0: Robust RMSEA <= 0.050	0.999
P-value H_0: Robust RMSEA >= 0.080	0.000

Standardized Root Mean Square Residual:

SRMR	0.017	0.017
------	-------	-------

Parameter Estimates:

Standard errors	Sandwich
Information bread	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
factor1 =~						
X1	0.578	0.039	14.876	0.000	0.578	0.566
X3	0.452	0.038	11.780	0.000	0.452	0.522
X5	0.659	0.041	16.022	0.000	0.659	0.682
X9	0.557	0.032	17.200	0.000	0.557	0.784
X10	0.613	0.035	17.707	0.000	0.613	0.705
X13	0.671	0.042	16.164	0.000	0.671	0.702
X14	0.559	0.039	14.279	0.000	0.559	0.376
X18	0.524	0.045	11.627	0.000	0.524	0.411
X21	0.555	0.043	13.021	0.000	0.555	0.669
X23	0.621	0.034	18.196	0.000	0.621	0.639
factor2 =~						
X3	-0.023	0.049	-0.466	0.641	-0.023	-0.027
X5	-0.098	0.069	-1.424	0.154	-0.098	-0.101
X9	-0.076	0.069	-1.093	0.274	-0.076	-0.107
X10	-0.108	0.075	-1.430	0.153	-0.108	-0.124
X13	0.218	0.072	3.034	0.002	0.218	0.228
X14	-0.011	0.082	-0.140	0.888	-0.011	-0.008
X18	0.306	0.073	4.205	0.000	0.306	0.240
X21	0.297	0.089	3.344	0.001	0.297	0.358
X23	0.161	0.069	2.322	0.020	0.161	0.166

Regressions:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
factor1 ~						
factor2	0.000				0.000	0.000

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.X1	0.709	0.065	10.958	0.000	0.709	0.680
.X3	0.545	0.043	12.668	0.000	0.545	0.727
.X5	0.489	0.053	9.306	0.000	0.489	0.525
.X9	0.189	0.020	9.463	0.000	0.189	0.375
.X10	0.369	0.043	8.665	0.000	0.369	0.488
.X13	0.417	0.048	8.621	0.000	0.417	0.456
.X14	1.896	0.078	24.415	0.000	1.896	0.858
.X18	1.258	0.101	12.423	0.000	1.258	0.774
.X21	0.291	0.041	7.040	0.000	0.291	0.424
.X23	0.534	0.051	10.465	0.000	0.534	0.564
.factor1	1.000				1.000	1.000
factor2	1.000				1.000	1.000

```
> summary(CFA_3factor.model, fit.measures = TRUE, standardized = TRUE)
lavaan 0.6-19 ended normally after 268 iterations
```

Estimator	ML
Optimization method	NLMINB
Number of model parameters	37
Number of observations	1333

Model Test User Model:	Standard	Scaled
Test Statistic	31.402	32.390
Degrees of freedom	18	18
P-value (Chi-square)	0.026	0.020
Scaling correction factor		0.969
Yuan-Bentler correction (Mplus variant)		

Model Test Baseline Model:		
Test statistic	4148.081	2238.585
Degrees of freedom	45	45
P-value	0.000	0.000
Scaling correction factor		1.853

User Model versus Baseline Model:		
Comparative Fit Index (CFI)	0.997	0.993
Tucker-Lewis Index (TLI)	0.992	0.984
Robust Comparative Fit Index (CFI)		0.997
Robust Tucker-Lewis Index (TLI)		0.991

Loglikelihood and Information Criteria:		
Loglikelihood user model (H0)	-16510.511	-16510.511
Scaling correction factor for the MLR correction		2.639
Loglikelihood unrestricted model (H1)	-16494.810	-16494.810
Scaling correction factor for the MLR correction		2.092
Akaike (AIC)	33095.022	33095.022
Bayesian (BIC)	33287.244	33287.244
Sample-size adjusted Bayesian (SABIC)	33169.711	33169.711

Root Mean Square Error of Approximation:

RMSEA	0.024	0.024
90 Percent confidence interval - lower	0.008	0.009
90 Percent confidence interval - upper	0.037	0.038
P-value H_0: RMSEA <= 0.050	1.000	0.999
P-value H_0: RMSEA >= 0.080	0.000	0.000

Robust RMSEA	0.024
90 Percent confidence interval - lower	0.010
90 Percent confidence interval - upper	0.037
P-value H_0: Robust RMSEA <= 0.050	1.000
P-value H_0: Robust RMSEA >= 0.080	0.000

Standardized Root Mean Square Residual:

SRMR	0.013	0.013
------	-------	-------

Parameter Estimates:

Standard errors	Sandwich
Information bread	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
factor1 =~						
X1	0.595	0.043	13.914	0.000	0.595	0.582
X3	0.464	0.043	10.767	0.000	0.464	0.535
X5	0.654	0.041	15.869	0.000	0.654	0.677
X9	0.527	0.043	12.124	0.000	0.527	0.742
X10	0.593	0.044	13.576	0.000	0.593	0.681
X13	0.648	0.048	13.616	0.000	0.648	0.677
X14	0.603	0.084	7.195	0.000	0.603	0.406
X18	0.512	0.053	9.601	0.000	0.512	0.401
X21	0.522	0.051	10.282	0.000	0.522	0.630
X23	0.631	0.046	13.752	0.000	0.631	0.649
factor2 =~						
X3	-0.009	0.091	-0.095	0.924	-0.009	-0.010
X5	-0.016	0.605	-0.027	0.978	-0.016	-0.017
X9	0.120	4.100	0.029	0.977	0.120	0.169
X10	0.009	0.647	0.014	0.989	0.009	0.010
X13	0.265	0.233	1.134	0.257	0.265	0.277
X14	-0.071	0.275	-0.256	0.798	-0.071	-0.047
X18	0.294	0.568	0.518	0.605	0.294	0.230
X21	0.381	0.496	0.767	0.443	0.381	0.459
X23	0.152	0.568	0.267	0.789	0.152	0.156

factor3 =~

X5	-0.081	0.154	-0.525	0.599	-0.081	-0.084
X9	-0.484	0.936	-0.517	0.605	-0.484	-0.681
X10	-0.093	0.404	-0.229	0.819	-0.093	-0.107
X13	0.022	2.198	0.010	0.992	0.022	0.023
X14	0.048	0.830	0.058	0.954	0.048	0.032
X18	0.079	2.210	0.036	0.971	0.079	0.062
X21	0.056	3.087	0.018	0.986	0.056	0.068
X23	0.075	1.096	0.069	0.945	0.075	0.077

Regressions:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
factor1 ~						
factor2	0.000				0.000	0.000
factor3	0.000				0.000	0.000
factor2 ~						
factor3	0.000				0.000	0.000

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.X1	0.689	0.069	9.938	0.000	0.689	0.661
.X3	0.535	0.053	10.047	0.000	0.535	0.714
.X5	0.498	0.057	8.754	0.000	0.498	0.535
.X9	-0.022	1.676	-0.013	0.990	-0.022	-0.043
.X10	0.397	0.059	6.732	0.000	0.397	0.524
.X13	0.425	0.049	8.640	0.000	0.425	0.464
.X14	1.839	0.163	11.309	0.000	1.839	0.832
.X18	1.271	0.099	12.827	0.000	1.271	0.782
.X21	0.267	0.048	5.611	0.000	0.267	0.388
.X23	0.519	0.054	9.535	0.000	0.519	0.549
.factor1	1.000				1.000	1.000
.factor2	1.000				1.000	1.000
factor3	1.000				1.000	1.000

CONCLUDING REMARKS

- Today we discussed the world of exploratory factor analysis and found the following:
 - PCA is what people typically run when they are after EFA
 - ML EFA is a better option to pick (likelihood based)
 - Constraints employed are hidden!
 - Rotations can break without you realizing they do
 - ML EFA can be shown to be equal to CFA for certain models
 - Overall, CFA is still your best bet